# Week 8

**1. What is the main goal of GraphX?**

A) Database management
B) Unify data-parallel and graph-parallel analytics
C) Replace Spark SQL
D) Replace HDFS

**Answer: B**

**Explanation**

**A. Database management (Incorrect):** GraphX is a graph analytics framework, not a database management system.

**B. Unify data-parallel and graph-parallel analytics (Correct):** GraphX is Spark's graph processing API that combines graph computation with Spark's data-parallel operations, allowing seamless analysis of both graph and non-graph data.

**C. Replace Spark SQL (Incorrect):** Spark SQL is used for structured data processing, while GraphX is specifically designed for graph analytics.

**D. Replace HDFS (Incorrect):** HDFS is a distributed storage system, whereas GraphX is a graph processing library built on top of Spark.

**2. Which phase is NOT part of the Graph-Parallel Pattern?**

A) Gather
B) Apply
C) Scatter
D) Shuffle Sort

**Answer: D**

**Explanation**

**A. Gather (Incorrect):** Gather is a valid Gather-Apply-Scatter phase where information is collected from neighboring vertices or edges.

**B. Apply (Incorrect):** Apply is a valid Gather-Apply-Scatter phase that updates a vertex's state using the gathered information.

**C. Scatter (Incorrect):** Scatter is a valid Gather-Apply-Scatter phase that propagates updated information to neighboring vertices for the next iteration.

**D. Shuffle Sort (Correct):** The Graph-Parallel Pattern in GraphX follows the Gather-Apply-Scatter model. Shuffle Sort is a MapReduce operation and is not one of the GAS phases.

**3. Which GraphX operator joins vertices and edges together?**

A) mapV()
B) subgraph()
C) triplets
D) reverse()

**Answer: C**

**Explanation**

**A. mapV() (Incorrect):** `mapV()` (or `mapVertices`) transforms vertex properties but does not join vertex and edge information together.

**B. subgraph() (Incorrect):** `subgraph()` creates a filtered graph by selecting specific vertices and edges; it does not combine their attributes.

**C. triplets (Correct):** The triplets operator provides a view that combines the source vertex, destination vertex, and edge attributes, enabling computations that require information from both vertices and edges.

**D. reverse() (Incorrect):** `reverse()` changes the direction of all edges in the graph but does not join vertices and edges into a single structure.

**4.In GraphX, a Property Graph consists of:**

A) Only vertices
B) Only edges
C) Vertices and edges with properties
D) Tables only

**Answer: C**

**Explanation**

**A. Only vertices (Incorrect):** A graph requires both vertices and edges; vertices alone cannot represent relationships.

**B. Only edges (Incorrect):** Edges represent connections, but without vertices there are no entities to connect.

**C. Vertices and edges with properties (Correct):** A Property Graph in GraphX contains both vertices and edges, and each can store associated attributes (properties) such as labels, weights, or metadata.

**D. Tables only (Incorrect):** Tables are used in relational databases, whereas GraphX represents data as graphs consisting of vertices and edges with properties.

**5. Which graph algorithm is commonly used to identify influential vertices?**

A) K-Means
B) PageRank
C) Apriori
D) Frequent Pattern Growth

**Answer: B**

**Explanation**

**A. K-Means (Incorrect):** K-Means is a clustering algorithm used to group similar data points, not to rank graph vertices.

**B. PageRank (Correct):** PageRank measures the importance of a vertex based on the number and quality of links pointing to it, making it useful for identifying influential nodes in a graph.

**C. Apriori (Incorrect):** Apriori is an association rule mining algorithm used for finding frequent itemsets in transactional data.

**D. Frequent Pattern Growth (Incorrect):** Frequent Pattern Growth is also a frequent pattern mining algorithm and is not designed to determine influential vertices in a graph.

**6.Triangle Counting is mainly used for:**

A) Sorting vertices
B) Community detection
C) Data compression
D) Replication

**Answer: B**

**Explanation**

**A. Sorting vertices (Incorrect):** Triangle Counting analyzes graph connectivity patterns and is not used for ordering or sorting vertices.

**B. Community detection (Correct):** Triangle Counting helps identify tightly connected groups of vertices in a graph. A high number of triangles often indicates strong community structure and social cohesion.

**C. Data compression (Incorrect):** Data compression reduces storage requirements, whereas Triangle Counting is a graph analytics technique.

**D. Replication (Incorrect):** Replication creates copies of data for fault tolerance and availability, which is unrelated to Triangle Counting.

**7. Which optimization reduces communication by avoiding unnecessary joins?**

A) Join Elimination
B) Data Duplication
C) Replication Factor
D) Hash Partitioning

**Answer: A**

**Explanation**

**A. Join Elimination (Correct):** Join Elimination removes unnecessary join operations, reducing data movement and communication overhead during graph processing.

**B. Data Duplication (Incorrect):** Data duplication creates additional copies of data and may increase storage and communication costs rather than reduce them.

**C. Replication Factor (Incorrect):** Replication Factor improves fault tolerance and availability but does not specifically eliminate join operations.

**D. Hash Partitioning (Incorrect):** Hash Partitioning distributes data across partitions efficiently, but it does not remove unnecessary joins from execution plans.

**8. Which GraphX feature helps accelerate joins and construct subgraphs efficiently?**

A) Bloom Filters
B) Indexing and Bitmaps
C) Commit Logs
D) MemTables

**Answer: B**

**Explanation**

**A. Bloom Filters (Incorrect):** Bloom Filters are probabilistic structures used for membership testing and are not the primary GraphX mechanism for accelerating joins and subgraph creation.

**B. Indexing and Bitmaps (Correct):** GraphX uses indexing and bitmap structures to efficiently locate vertices and edges, speeding up join operations and subgraph construction.

**C. Commit Logs (Incorrect):** Commit Logs are used in systems like Cassandra for durable writes and have no role in GraphX graph processing.

**D. MemTables (Incorrect):** MemTables are in-memory write buffers used in Cassandra and are unrelated to GraphX optimizations.

**9. Which statement best describes GraphX?**

A) Graph-only processing framework
B) Batch-only processing framework
C) Unified analytics framework for graph and data processing
D) SQL replacement system

**Answer: C**

**Explanation**

**A. Graph-only processing framework (Incorrect):** GraphX is not limited to graph processing; it leverages Spark's broader data processing capabilities.

**B. Batch-only processing framework (Incorrect):** While GraphX runs on Spark, its purpose is not merely batch processing but the integration of graph and data analytics.

**C. Unified analytics framework for graph and data processing (Correct):** GraphX integrates graph-parallel computation with Spark's data-parallel processing, enabling users to perform graph analytics and general data analytics within the same framework.

**D. SQL replacement system (Incorrect):** GraphX is designed for graph computation and analytics, whereas Spark SQL is used for structured data querying and processing.

**10.Which GraphX optimization reuses routing tables, edge adjacency information, and indices across graphs and subgraphs?**

A) Substantial Index and Data Reuse
B) Active Edge Indexing
C) Join Elimination
D) Pregel Messaging

**Answer: A**

**Explanation**

**A. Substantial Index and Data Reuse (Correct):** GraphX reuses routing tables, edge adjacency structures, and vertex indices across derived graphs and subgraphs, reducing memory usage and improving performance.

**B. Active Edge Indexing (Incorrect):** Active Edge Indexing helps efficiently identify active edges during computation but does not focus on reusing graph structures across subgraphs.

**C. Join Elimination (Incorrect):** Join Elimination reduces communication overhead by avoiding unnecessary joins, but it does not reuse routing tables or indices.

**D. Pregel Messaging (Incorrect):** Pregel Messaging is a vertex-centric communication model used for iterative graph algorithms and is unrelated to index and data reuse.